



Advanced Time Series Analysis and Forecasting

Chapter 11: Spectral and wavelet analysis



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Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ARMA: AutoRegressive Moving Average (model)
- ▣ BC: Breitung–Candelon (frequency-domain causality test)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BK: Baxter–King (band-pass filter)
- ▣ BUX: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- ▣ CF: Christiano–Fitzgerald (band-pass filter)
- ▣ CFR: Croux, Forni and Reichlin (2001)
- ▣ COI: Cone Of Influence (of a wavelet transform)
- ▣ CWT: Continuous Wavelet Transform
- ▣ DCC-GARCH: Dynamic Conditional Correlation GARCH
- ▣ DFT: Discrete Fourier Transform
- ▣ DPSS: Discrete Prolate Spheroidal Sequences (Slepian tapers)
- ▣ DWT: Discrete Wavelet Transform
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ FFT: Fast Fourier Transform
- ▣ FRED: Federal Reserve Economic Data
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GDP: Gross Domestic Product
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HICP: Harmonised Index of Consumer Prices
- ▣ HP: Hodrick–Prescott (filter)
- ▣ LA: Least Asymmetric (Daubechies wavelet filter)
- ▣ LLM: Large Language Model

Acronyms used in this material (2/2)

- MA: Moving Average
- MODWT: Maximal Overlap Discrete Wavelet Transform
- MRA: MultiResolution Analysis
- MSE: Mean Squared Error
- PX: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- QS: Quadratic Spectral (kernel)
- RMSE: Root Mean Squared Error
- SSA: Singular Spectrum Analysis
- SVD: Singular Value Decomposition
- TSA: Time Series Analysis
- US: United States
- VAR: Vector AutoRegression
- VARMA: Vector AutoRegressive Moving Average (model)
- WIG20: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Today's question and route

- **Question:** how is the variance of a time series, and the co-movement of two series, distributed over frequencies, and how does this distribution change over time?
 - ▶ cycles of different lengths answer different economic questions: the business cycle, seasonality, the reaction of markets within a week
- **Route** of the chapter
 - ▶ the spectral representation; estimation theory: consistency, bandwidth, multitaper
 - ▶ two series: coherence, phase, dynamic correlation, frequency-domain causality
 - ▶ band-pass filters and Hamilton's critique of the HP filter; Romania and the euro area
 - ▶ evolutionary spectra; wavelets: MODWT, variance and correlation by scale, wavelet coherence
- We build on [TSA, Chapter 12](#) (periodogram, smoothing, Fisher's test, coherence), [Chapter 6](#) (state space) and [Chapter 10](#) (the spectral pole of long memory); Seminar 11 comes before this lecture

Learning outcomes

- ▣ State the spectral representation of a stationary process and derive the spectrum of a filtered process
- ▣ Derive the bias and variance of lag-window estimators, choose the bandwidth, and build multitaper estimates with valid confidence bands
- ▣ Estimate and test coherence, phase, gain and dynamic correlation, and test Granger causality at a given frequency
- ▣ Evaluate trend–cycle filters by their gain, and measure business-cycle synchronisation
- ▣ Decompose variance and correlation by scale with the MODWT and test wavelet coherence by Monte Carlo

Reading, data and tools

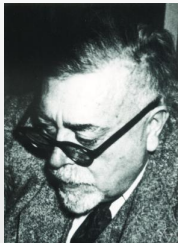
- ▣ Spectral theory and estimation: Brockwell and Davis (1991); Percival and Walden (1993); Shumway and Stoffer (2017); Thomson (1982)
 - ▶ Wavelets: Percival and Walden (2000); Torrence and Compo (1998); Grinsted, Moore and Jevrejeva (2004); Aguiar-Conraria and Soares (2014)
 - ▶ Business cycles: Baxter and King (1999); Christiano and Fitzgerald (2003); Hamilton (2018); Croux, Forni and Reichlin (2001)
 - ▶ Survey of forecasting practice: Petropoulos et al. (2022)
- ▣ Python Quantlets of this chapter: [Quantlets/Ch_11](#)
 - ▶ spectra, multitaper, cross-spectra, Geweke and Breitung–Candelon, filters, MODWT, Morlet transform and wavelet coherence written in `numpy/scipy`; `PyWavelets` gives the wavelet filters
- ▣ Lecture notebook: [open in Google Colab](#)

Data used in this chapter

Series	Source	Sample
Real GDP: Romania, euro area, Poland, Hungary, Czechia	Eurostat, chain-linked volumes, seasonally and calendar adjusted, quarterly	1995Q1 – 2026Q2
Industrial production: Romania, euro area	Eurostat, monthly, adjusted and unadjusted	January 2000 – July 2026
HICP annual inflation: Romania, Poland, euro area	Eurostat, monthly	January 2001 – August 2026
BET, DAX, S&P 500	EODHD daily closes; weekly returns from Friday closes	2000 – 2026
Brent crude oil	FRED (US Energy Information Administration), daily	2000 – 2026

- Growth rates and returns are 100 times log differences; cycles are in % of trend; markets are aligned on common trading days

From harmonic analysis to time series



Norbert Wiener

Photo: Konrad Jacobs, MFO; CC BY-SA 2.0 de;

[Wikimedia Commons](#)

- 1930: Wiener's generalised harmonic analysis; 1934: Khinchin; autocovariance and spectrum are a Fourier pair
- 1942: Cramér's representation of a stationary process as a random superposition of sinusoids
- 1961–1966: lag windows [Parzen \(1961\)](#); the typical spectral shape of economic variables [Granger \(1966\)](#)
- 1982: multitaper estimation [Thomson \(1982\)](#); frequency-domain causality [Geweke \(1982\)](#)
- 1984–1989: wavelets [Grossmann and Morlet \(1984\)](#), [Daubechies \(1988\)](#), [Mallat \(1989\)](#); 1998: the practical guide [Torrence and Compo \(1998\)](#)

Known from TSA and new here

- Known (TSA, Chapter 12): Fourier frequencies, the spectrum of ARMA models, the periodogram and its χ^2_2 law, Daniell smoothing, tapering, Fisher's test, a first look at coherence and wavelets
- New: the research toolkit
 - ▶ asymptotic theory of spectral estimators: bias, variance, optimal bandwidth, valid confidence bands, multitaper
 - ▶ inference on co-movement: coherence thresholds, phase errors, dynamic correlation, causality by frequency
 - ▶ filters judged by their gain; time–frequency analysis with significance tests that control false discoveries
- Case studies: Thomson (1982), Croux–Forni–Reichlin (2001), Breitung–Candelon (2006), Baxter–King (1999), Cogley–Nason (1995), Hamilton (2018), Torrence–Compo (1998), Grinsted et al. (2004), with the specifications of the papers, on our data

Autocovariance and spectral distribution (1/2)

- **Herglotz Brockwell and Davis (1991)**: $\gamma(\cdot)$ is the autocovariance of a stationary process if and only if

$$\gamma(h) = \int_{(-\pi, \pi]} e^{i\omega h} dF(\omega)$$

- ▶ $\gamma(h) = \text{Cov}(X_{t+h}, X_t)$; ω : angular frequency in radians per period; F : the **spectral distribution**, non-decreasing, right-continuous and bounded

- If $\sum_h |\gamma(h)| < \infty$, F has a density, the **spectral density**

$$dF(\omega) = f(\omega) d\omega, \quad f(\omega) = \frac{1}{2\pi} \sum_h \gamma(h) e^{-i\omega h}$$

Autocovariance and spectral distribution (2/2)

- At $h = 0$: the spectrum is a **decomposition of variance by frequency**

$$\gamma(0) = \text{Var}(X_t) = \int_{-\pi}^{\pi} f(\omega) d\omega$$

- ▶ $f(\omega) d\omega$: the variance contributed by cycles with frequency near ω ; period $2\pi/\omega$
- ▶ quarterly data: the business-cycle band of 6–32 quarters is $\omega \in [\pi/16, \pi/3]$
- A jump of F at $\pm\omega_0$ is a deterministic cycle (a **line spectrum**); long memory is a pole of f at $\omega = 0$ ([Chapter 10](#))

Cramér's representation



Harald Cramér, 1951

Photo: unknown photographer, Svenska
Dagbladet (1951); public domain; Wikimedia
Commons

- Every zero-mean stationary process is a sum of sinusoids with random amplitudes [Brockwell and Davis \(1991\)](#)

$$X_t = \int_{(-\pi, \pi]} e^{i\omega t} dZ(\omega)$$

- ▶ $dZ(\omega)$: the random amplitude at frequency ω ; Z has **orthogonal increments**: $\mathbb{E}[dZ(\omega)\overline{dZ(\lambda)}] = 0$ for $\omega \neq \lambda$, $\mathbb{E}|dZ(\omega)|^2 = dF(\omega)$
- Random amplitudes at different frequencies are uncorrelated: the frequency components can be studied one by one
- The discrete Fourier transform (DFT) $d(\omega_j) = n^{-1/2} \sum_t x_t e^{-i\omega_j t}$ at $\omega_j = 2\pi j/n$ is the sample analogue of $dZ(\omega_j)$: nearly uncorrelated across Fourier frequencies

Linear filters in the frequency domain

- A linear filter multiplies each frequency component by the **transfer function** $A(\omega)$

$$Y_t = \sum_j a_j X_{t-j} \Rightarrow dZ_Y(\omega) = A(\omega) dZ_X(\omega), \quad A(\omega) = \sum_j a_j e^{-i\omega j}$$

$$f_Y(\omega) = |A(\omega)|^2 f_X(\omega)$$

- ▶ a_j : filter weights, $\sum_j |a_j| < \infty$; $|A(\omega)|$: the **gain**; $\arg A(\omega)$: the **phase shift**
- ▶ symmetric weights $a_j = a_{-j}$: $A(\omega)$ is real, no phase shift; the property every trend-cycle filter wants
- First difference: $|1 - e^{-i\omega}|^2 = 2(1 - \cos \omega)$, zero at $\omega = 0$
 - ▶ differencing removes the long cycles and amplifies the short ones
- Slutsky effect: a moving sum of white noise has a peak in its spectrum, i.e. **cycles created by the filter**; the HP critique of Section 4 is the same phenomenon

The spectral density matrix (1/2)

- Vector process X_t of dimension k : the Fourier transform of the autocovariance matrices

$$f(\omega) = \frac{1}{2\pi} \sum_h \Gamma(h) e^{-i\omega h}, \quad \Gamma(h) = \text{Cov}(X_{t+h}, X_t)$$

- ▶ $f(\omega)$: a $k \times k$ Hermitian ($f = f^*$, the conjugate transpose) non-negative definite matrix at each frequency

- VARMA $\Phi(L)X_t = \Theta(L)\varepsilon_t$: the multivariate analogue of the ARMA spectrum

$$f(\omega) = \frac{1}{2\pi} \Phi(e^{-i\omega})^{-1} \Theta(e^{-i\omega}) \Sigma \Theta(e^{-i\omega})^* \Phi(e^{-i\omega})^{-*}$$

- ▶ Φ, Θ : matrix AR and MA polynomials; Σ : covariance of the innovations ε_t ; $^{-*}$: inverse of the conjugate transpose

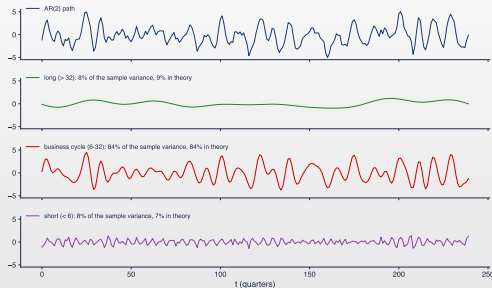
The spectral density matrix (2/2): cross-spectra

- Off-diagonal elements: the **cross-spectrum**, complex-valued

$$f_{xy}(\omega) = c_{xy}(\omega) - i q_{xy}(\omega)$$

- ▶ c_{xy} : co-spectrum, even in ω (contemporaneous co-movement), $\int c_{xy} = \text{Cov}(X_t, Y_t)$
- ▶ q_{xy} : quadrature spectrum, odd in ω (lead-lag between the series)
- Section 2 estimates the diagonal, Section 3 the off-diagonal, Section 4 factorises f to measure causality

Variance by frequency band



- An AR(2) path, $X_t = 1.3X_{t-1} - 0.7X_{t-2} + \varepsilon_t$, 240 “quarters”, split by the DFT into three bands; shares of variance in the sample and $2 \int_{\text{band}} f / \gamma(0)$ in theory

Interpreting the band decomposition

- ▣ The spectral peak is at a period of 9.5: 84% of the variance lies in the band of 6–32 periods (sample: 84%)
- ▣ Long and short cycles carry 9% and 7% in theory: the three components are uncorrelated by Cramér's theorem, so the shares add up
- ▣ The “business cycle” of an AR(2) is a stochastic pseudo-cycle: irregular amplitude and length, no line in the spectrum
- ▣ Band-pass filters (Section 4) estimate exactly this middle component from a finite sample

Recap: The spectral representation

- ▣ The spectrum decomposes variance by frequency; Cramér: uncorrelated random amplitudes at each frequency
- ▣ A linear filter multiplies the spectrum by its squared gain; symmetric filters do not shift phase
- ▣ The spectral matrix contains cross-spectra: co-movement and lead-lag by frequency

Why the periodogram is not enough

- For a linear process the periodogram ordinates are asymptotically scaled χ_2^2 variables [Brockwell and Davis \(1991\)](#)

$$X_t = \sum_j \psi_j \varepsilon_{t-j} \Rightarrow I(\omega_j) \Rightarrow f(\omega_j) \frac{\chi_2^2}{2}$$

- ▶ ψ_j : weights with $\sum_j |j|^{1/2} |\psi_j| < \infty$; ε_t : i.i.d. innovations; $f > 0$; asymptotically independent over fixed Fourier frequencies ω_j
- ▶ $\mathbb{E}I(\omega_j) \rightarrow f(\omega_j)$ but $\text{Var}I(\omega_j) \rightarrow f(\omega_j)^2$: **inconsistent**, the variance does not fall with n
- Leakage: the mean periodogram is the true spectrum smoothed by the Fejér kernel F_n

$$\mathbb{E}I(\omega) = \int F_n(\omega - \lambda) f(\lambda) d\lambda$$

- ▶ the side lobes of F_n decay only like $1/(n\omega^2)$, so power **leaks** from strong to weak frequencies
- Two remedies: **average** neighbouring frequencies (consistency) and **taper** the data (bias); multitaper does both

Lag-window estimators (1/2)

- Down-weight the noisy sample autocovariances at long lags [Parzen \(1961\)](#)

$$\hat{f}(\omega) = \frac{1}{2\pi} \sum_{|h| < n} k(h/M) \hat{\gamma}(h) e^{-i\omega h}$$

- ▶ $\hat{\gamma}(h)$: sample autocovariance; k : the **lag window**, $k(0) = 1$, decreasing in $|u|$; M : the truncation (bandwidth) parameter

- Equivalently: a weighted average of the periodogram with the **spectral window** W_M

$$\hat{f}(\omega) = \int W_M(\omega - \lambda) I(\lambda) d\lambda, \quad W_M(\omega) = \frac{1}{2\pi} \sum_h k(h/M) e^{-i\omega h}$$

- ▶ a large M gives a narrow W_M : little smoothing

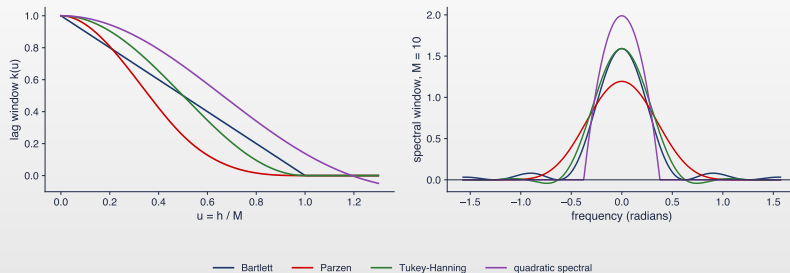
Lag-window estimators (2/2): the kernels

- Bartlett $k(u) = 1 - |u|$; Parzen; Tukey–Hanning; quadratic spectral (QS): the same kernels as the HAC estimators of [Chapter 0](#) at $\omega = 0$
 - ▶ a non-negative W_M guarantees $\hat{f} \geq 0$ (Bartlett, Parzen, QS); Tukey–Hanning can go negative
- Characteristic exponent q : how flat the kernel is at zero

$$k_q = \lim_{u \rightarrow 0} \frac{1 - k(u)}{|u|^q}$$

- ▶ Bartlett $q = 1$, $k_1 = 1$; Parzen $q = 2$, $k_2 = 6$; QS $q = 2$, $k_2 = 1.42$; a larger q means smaller bias (next slides)

Lag windows and spectral windows



- Left: $k(u)$; right: the spectral window $W_M(\omega)$ for $M = 10$, the weights that each estimator gives to the periodogram around ω

Interpreting the windows

- ▣ A narrow W_M means low bias and high variance; Parzen is the widest at $M = 10$ (its $\int k^2 = 0.54$ is the smallest), QS the narrowest
- ▣ Bartlett has side lobes (leakage through the window itself); Tukey–Hanning has negative weights
- ▣ Comparing kernels at the same M is misleading: compare them at the same variance (the same equivalent degrees of freedom)
- ▣ The QS kernel is optimal in MSE among kernels with non-negative estimates [Andrews \(1991\)](#)

Bias, variance and the optimal bandwidth (1/2)

- If $M \rightarrow \infty$ and $M/n \rightarrow 0$: $\hat{f}(\omega) \rightarrow_p f(\omega)$ (**consistency**) [Parzen \(1961\)](#); variance and bias

$$\text{Var}\hat{f}(\omega) \approx \frac{M}{n} f(\omega)^2 \int k^2(u) du, \quad \mathbb{E}\hat{f}(\omega) - f(\omega) \approx -k_q M^{-q} f^{(q)}(\omega)$$

- ▶ the variance grows with M (doubled at $\omega = 0, \pm\pi$); the bias falls with M
- ▶ $f^{(q)}(\omega) = \frac{1}{2\pi} \sum_h |h|^q \gamma(h) e^{-i\omega h}$: a generalised q -th derivative of f , large where the spectrum is curved (peaks)

Bias, variance and the optimal bandwidth (2/2)

- Minimising MSE = squared bias + variance over M

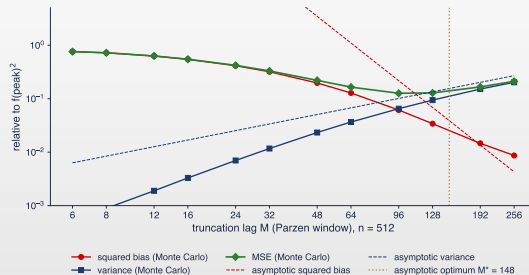
$$M^* = \left(\frac{q k_q^2 f^{(q)}(\omega)^2 n}{f(\omega)^2 \int k^2} \right)^{1/(2q+1)} \propto n^{1/(2q+1)}, \quad \text{MSE} \propto n^{-2q/(2q+1)}$$

- ▶ $q = 2$ kernels converge faster ($n^{-4/5}$) than Bartlett ($n^{-2/3}$)
- Inference: a scaled χ^2 with equivalent degrees of freedom ν

$$\frac{\nu \hat{f}(\omega)}{f(\omega)} \approx \chi_\nu^2, \quad \nu = \frac{2n}{M \int k^2}$$

- ▶ the confidence band $[\nu \hat{f} / \chi_{\nu, 0.975}^2, \nu \hat{f} / \chi_{\nu, 0.025}^2]$ has constant width in log scale

The bias–variance trade-off at a spectral peak



- Parzen estimator at the peak (period 11.1) of $X_t = 1.6X_{t-1} - 0.9X_{t-2} + \varepsilon_t$, $n = 512$, 400 simulations; curves divided by $f(\text{peak})^2$; dashed: the asymptotic formulas

Interpreting the trade-off

- ▣ With $M = 12$ the relative RMSE is 0.79, almost all of it bias (0.79): heavy smoothing flattens the peak
- ▣ The Monte Carlo optimum is $M = 96$ (relative RMSE 0.36); the asymptotic formula gives $M^* = 148$: same order, but the formula ignores higher-order bias terms
- ▣ The asymptotic variance is accurate only once the window is narrower than the peak; the asymptotic bias explodes for small M
- ▣ A sharp spectral peak needs a much larger M than the HAC rule for $\omega = 0$ would suggest: bandwidth is local

Choosing the bandwidth

- ▣ **Plug-in Andrews (1991)**: approximate $f^{(q)}/f$ by a parametric (AR(1)) model, then use M^*

$$M = 1.1447 (\hat{\alpha}(1) n)^{1/3} \text{ (Bartlett)}, \quad M = 1.3221 (\hat{\alpha}(2) n)^{1/5} \text{ (QS)}, \quad \omega = 0$$

- ▶ $\hat{\alpha}(q)$: the ratio $(f^{(q)}(0)/f(0))^2$ computed from the fitted AR(1), e.g. $\hat{\alpha}(2) = 4\hat{\rho}^2/(1 - \hat{\rho})^4$ ($\hat{\rho}$: AR coefficient)
- ▶ good for long-run variances ([Chapter 0](#)); poor near sharp peaks, where the AR(1) approximation is wrong

- ▣ **Cross-validation Hurvich (1985)**: minimise over M a leave-one-out Whittle criterion

$$CV(M) = \sum_j \left[\log \hat{f}_{-j}(\omega_j) + \frac{I(\omega_j)}{\hat{f}_{-j}(\omega_j)} \right]$$

- ▶ $\hat{f}_{-j}$: the smoothed estimate computed without the ordinate $I(\omega_j)$

- ▣ **Resolution**: two peaks closer than the bandwidth (about $1/M$ cycles) merge; decide the resolution you need before looking at the data
- ▶ report the estimate for two or three bandwidths: conclusions that depend on M are not conclusions

Multitaper estimation (1/2): Slepian tapers

- ▣ **Slepian (DPSS) tapers** $v^{(k)}$, $k = 0, \dots, K - 1$ Thomson (1982)
 - ▶ orthonormal weight sequences of length n that maximise the share λ_k of their spectral energy inside $[-W, W]$; W : half-bandwidth in cycles
 - ▶ about $2nW$ of them have $\lambda_k \approx 1$; usually $K = 2NW - 1$ with $NW = nW$ (e.g. $NW = 4$, $K = 7$)
- ▣ Eigenspectra: one tapered periodogram per taper; the estimate is their average

$$\hat{S}_k(\omega) = \frac{1}{2\pi} \left| \sum_t v_t^{(k)} x_t e^{-i\omega t} \right|^2, \quad \hat{f}^{\text{mt}}(\omega) = \frac{1}{K} \sum_{k=0}^{K-1} \hat{S}_k(\omega)$$

- ▶ the $\hat{S}_k$ are nearly uncorrelated: $2K\hat{f}^{\text{mt}}/f \approx \chi_{2K}^2$; variance f^2/K , bias controlled by the taper concentration

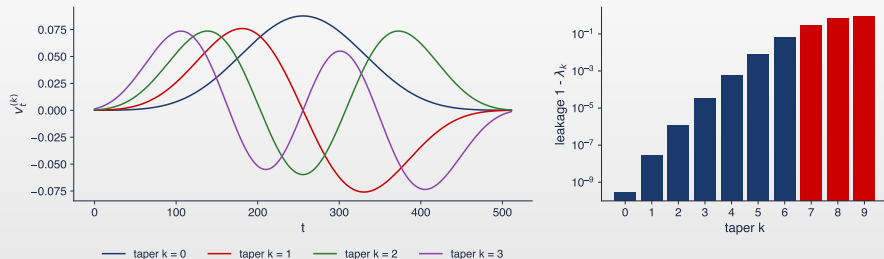
Multitaper estimation (2/2): adaptive weights

- ▣ **Adaptive weights** Percival and Walden (1993): down-weight leaky tapers where f is small

$$d_k(\omega) = \frac{\sqrt{\lambda_k} f(\omega)}{\lambda_k f(\omega) + (1 - \lambda_k) \sigma^2}$$

- ▶ σ^2 : the variance of the series; $1 - \lambda_k$: the leakage of taper k ; the weights depend on the unknown f , so they are iterated from $\hat{f}^{\text{mt}}$
- ▣ Alternatives with the same logic: sine tapers Riedel and Sidorenko (1995); Welch's averaging of tapered segments Welch (1967)

Slepian tapers



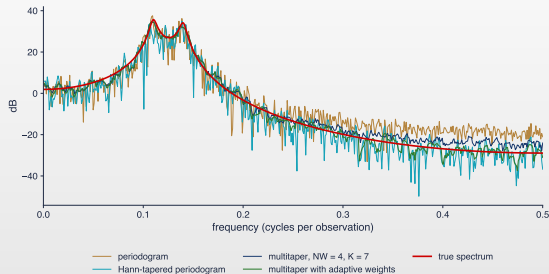
□ $n = 512$, $NW = 4$: the first four tapers and the leakage $1 - \lambda_k$ of the first ten (blue: $k < 2NW - 1$)

ATS_ch11_spectral_estimation

Interpreting the tapers

- ▣ Taper k has k zero crossings: higher tapers weight the ends of the sample, so together they use all the data
- ▣ Leakage grows fast with k : $\lambda_6 = 0.937$, but $\lambda_7 = 0.70$ and $\lambda_8 = 0.30$; beyond $2NW - 1$ the tapers are useless
- ▣ The single Hann taper throws away the ends; multitapering recovers that information as extra degrees of freedom
- ▣ NW is the only tuning choice: it fixes the resolution $2W = 2NW/n$ and the variance f^2/K together

Leakage on a spectrum with a high dynamic range

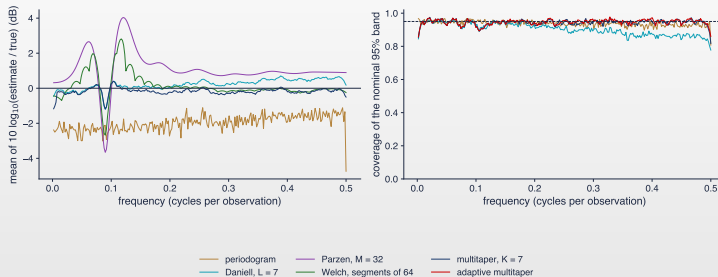


- The AR(4) of Percival and Walden (1993), $n = 1024$, range of 65 dB between peak and trough; one simulated path (median leakage of 21)

Interpreting leakage

- ▣ Mean bias at frequencies above 0.3 cycles (200 simulations): periodogram +11.2 dB, Hann taper -2.5 dB, multitaper +6.3 dB, adaptive multitaper -0.8 dB
- ▣ The raw periodogram fills the trough with power leaked from the peaks: averaging it more would not help, the bias is in each ordinate
- ▣ With equal weights the leaky high-order tapers still contaminate the trough; the adaptive weights remove it
- ▣ Economic data rarely have 60 dB ranges, but spectra of levels with a unit-root-like peak at zero do: taper or prewhiten before estimating

Five estimators in a Monte Carlo

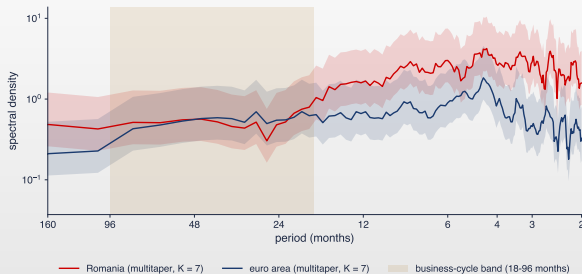


- $X_t = 1.6X_{t-1} - 0.9X_{t-2} + \varepsilon_t$, $n = 512$, 400 simulations: mean log-bias by frequency and coverage of the nominal 95% χ^2 bands

Interpreting the Monte Carlo

- Standard deviation of $10 \log_{10}(\hat{f}/f)$: periodogram 5.6 dB, Daniell 1.8, multitaper 1.7, Parzen 0.9, Welch 1.1
- At the peak: Parzen -2.3 dB, Welch -1.4 dB, multitaper -0.5 dB; the periodogram's -2.0 dB is the mean of $\log \chi^2_2/2$, not a bias in levels
- Coverage of the 95% band: multitaper 94.3%, adaptive 94.6%, Daniell 90.3% (its band ignores the bias on the slopes)
- Multitaper is the only estimator here with small bias, small variance and honest bands at once

Industrial production: Romania and the euro area



- Monthly growth of industrial production (seasonally and calendar adjusted), February 2000 – July 2026, 318 months; multitaper $NW = 4$ with 95% bands

Interpreting the industrial production spectra

- Monthly growth is dominated by short periods: cycles under 6 months carry 77% of the variance in Romania and 69% in the euro area
- The business-cycle band (18–96 months) holds 2.1% (Romania) and 7.3% (euro area): growth rates hide cycles that levels show (the first-difference gain)
- Romania is more volatile (standard deviation 3.61 against 1.98), and the excess is at high frequencies, where the bands do not overlap: noise, revisions and residual calendar effects
- At periods above 12 months the two spectra cannot be told apart within the bands

Lines in the spectrum: Thomson's F test (1/2)

- Model: a sinusoid plus noise with a smooth spectrum [Thomson \(1982\)](#)

$$x_t = \mu \cos(\omega_0 t + \phi) + e_t$$

- ▶ μ : amplitude; ω_0 : frequency of the line; ϕ : phase; e_t : stationary noise
- ▶ under a line, each tapered DFT is $Y_k(\omega_0) \approx \mu U_k(0)$, with $U_k(0) = \sum_t v_t^{(k)}$ the sum of taper k
- A regression of the eigencoefficients Y_k on the taper sums $U_k(0)$, frequency by frequency: it separates a deterministic line from a stochastic peak, which Fisher's test (TSA) cannot

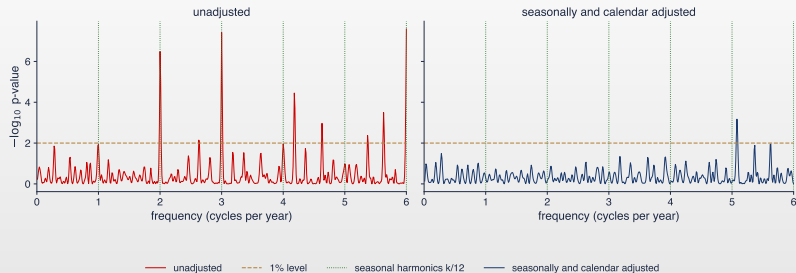
Lines in the spectrum: Thomson's F test (2/2)

- Estimated amplitude and the F statistic: explained against residual variation

$$\hat{\mu}(\omega) = \frac{\sum_k U_k(0) Y_k(\omega)}{\sum_k U_k(0)^2}, \quad F(\omega) = \frac{(K-1) |\hat{\mu}|^2 \sum_k U_k(0)^2}{\sum_k |Y_k - \hat{\mu} U_k(0)|^2} \sim F_{2, 2K-2}$$

- ▶ a large $F(\omega)$: a line at ω ; K : number of tapers
- Application: residual seasonality is a set of lines at $k/12$ cycles per month; trading-day effects are lines at 0.348 and 0.432 cycles per month
- Testing at hundreds of frequencies: expect about 1% false rejections at the 1% level; test only pre-specified frequencies

Residual seasonality in Romanian industrial production



- Harmonic F test ($NW = 4$, $K = 7$) on monthly growth, unadjusted and adjusted series; dotted lines: the six seasonal frequencies

Interpreting the F test

- Unadjusted: 3 of 6 seasonal harmonics are lines at 1% ($p < 0.0001$ and < 0.0001 at 2 and 3 cycles per year); the annual harmonic is borderline ($p = 0.012$)
- A strong line at 4.18 cycles per year ($p < 0.0001$): the trading-day frequency, a calendar effect, not seasonality
- Adjusted series: none of the six harmonics is significant, and there is no line at the trading-day frequency ($p = 0.22$): the adjustment passes this check
- The annual harmonic is weak because Romanian seasonality is not a pure sinusoid: its power sits at the higher harmonics

Recap: Spectral estimation theory

- The periodogram is unbiased in the limit but inconsistent and leaky; smoothing buys consistency, tapering buys low bias
- Bandwidth is a bias–variance choice with $M^* \propto n^{1/(2q+1)}$; peaks need narrow windows
- Multitaper: χ^2_{2K} bands that hold, adaptive weights against leakage, and an F test for lines

Coherence, phase and gain (1/2): coherence

- ▣ **Squared coherence:** the R^2 of the regression of $dZ_y(\omega)$ on $dZ_x(\omega)$

$$\kappa_{xy}^2(\omega) = \frac{|f_{xy}(\omega)|^2}{f_x(\omega)f_y(\omega)} \in [0, 1]$$

- ▶ f_x, f_y : the two spectra; f_{xy} : the cross-spectrum; $\kappa^2 = 1$: y is an exact linear filter of x at frequency ω ; $\kappa^2 = 0$: no linear relation at that frequency
 - ▶ invariant to filtering each series by an invertible filter: coherence does not depend on how the series are transformed (▶ [Appendix](#))
- ▣ **Gain** $|f_{xy}(\omega)|/f_x(\omega)$: the regression coefficient of y on x at frequency ω

Coherence, phase and gain (2/2): phase

▣ **Phase:** the angle of the cross-spectrum, with $\gamma_{xy}(h) = \text{Cov}(x_{t+h}, y_t)$

$$\phi_{xy}(\omega) = \arg f_{xy}(\omega)$$

- ▶ example: if $y_t = x_{t-d}$, then $f_{xy} = e^{i\omega d} f_x$, phase $\omega d > 0$: x leads by ϕ/ω periods
- ▶ the phase is defined modulo 2π : a lead of d and a lag of $2\pi/\omega - d$ are the same phase

▣ Interpret the phase only where coherence is significant

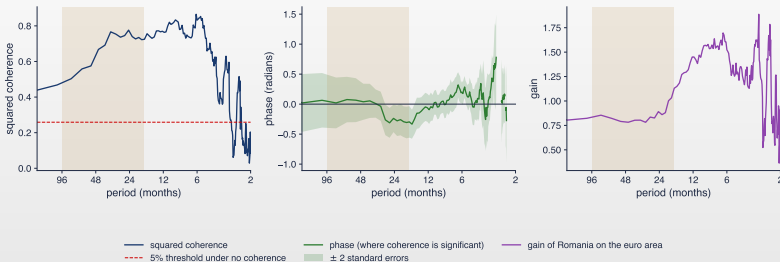
Inference on coherence and phase

- The raw cross-periodogram gives $\hat{\kappa}^2 \equiv 1$ at every frequency: coherence **must** be smoothed (here: averaged over K tapers)

$$\Pr(\hat{\kappa}^2 > c \mid \kappa^2 = 0) = (1 - c)^{L-1} \Rightarrow c_{0.05} = 1 - 0.05^{1/(L-1)}$$

- ▶ L : number of independent ordinates averaged ($L = K$ for multitaper); c : threshold; $\hat{\kappa}^2$ above $c_{0.05}$ is significant at 5%
- $\hat{\kappa}^2$ is biased upwards when κ^2 is small; confidence intervals via the Fisher transform $\tanh^{-1} \hat{\kappa}$ [Shumway and Stoffer \(2017\)](#)
 - ▶ phase: $\text{se}(\hat{\phi}) \approx \sqrt{(1/\hat{\kappa}^2 - 1)/(2L)}$; it explodes as coherence falls
- Misalignment bias: a long delay d rotates the phase within the smoothing band and lowers $\hat{\kappa}^2$; align the series (or prewhiten) first

Romania and the euro area: coherence, phase and gain



□ Monthly industrial production growth, euro area (x) and Romania (y); multitaper $NW = 6$, $K = 11$; shaded: 18–96 months

Q.ATS_ch11_cross_spectrum

Interpreting the cross-spectrum

- ▣ Coherence averages 0.69 in the business-cycle band and 0.47 below 12 months; the 5% threshold is 0.26: significant at 100% of business-cycle frequencies and 71% of short ones
- ▣ The phase is near zero: the median implied lead is 0.5 months, inside two standard errors: Romanian industry moves with the euro area within the month
- ▣ Gain 0.85 at business-cycle frequencies, 1.24 at short ones: Romania amplifies euro-area shocks only at high frequencies
- ▣ The simple correlation (0.66) mixes these bands; the coherence separates where co-movement is strong from where it is noise

Dynamic correlation

- Croux, Forni and Reichlin (2001): the correlation of the in-phase (real) components at frequency ω

$$\rho_{xy}(\omega) = \frac{c_{xy}(\omega)}{\sqrt{f_x(\omega)f_y(\omega)}} \in [-1, 1]$$

$$\kappa^2 = \rho^2 + \left(\frac{q_{xy}}{\sqrt{f_x f_y}} \right)^2$$

- ▶ c_{xy} , q_{xy} : co-spectrum and quadrature spectrum; unlike coherence, ρ_{xy} has a **sign** and is not increased by out-of-phase movement
- Over a band Λ of frequencies

$$\rho_{xy}(\Lambda) = \frac{\int_{\Lambda} c_{xy}}{\sqrt{\int_{\Lambda} f_x \int_{\Lambda} f_y}}$$

- ▶ over $[0, \pi]$ it is the ordinary correlation; over a band it equals the correlation of the band-pass filtered series (Section 4)
- **Cohesion**: a weighted average of pairwise $\rho_{xy}(\omega)$ inside a group of countries; CFR use it for euro-area business cycles

Granger causality by frequency (1/2): the Geweke measure

- Geweke (1982): in a bivariate VAR, normalise the innovations so that x 's innovation is uncorrelated with the transformed y innovation; then

$$f_x(\omega) = \frac{1}{2\pi} \left(|\tilde{H}_{xx}(\omega)|^2 \tilde{\sigma}_{xx} + |\tilde{H}_{xy}(\omega)|^2 \tilde{\sigma}_{yy} \right)$$

- ▶ $\tilde{H}$: the transfer function (moving-average representation) of the normalised VAR; $\tilde{\sigma}_{xx}$, $\tilde{\sigma}_{yy}$: variances of the two orthogonal innovations

- The share of $f_x(\omega)$ that comes from y

$$M_{y \rightarrow x}(\omega) = \ln \frac{2\pi f_x(\omega)}{|\tilde{H}_{xx}(\omega)|^2 \tilde{\sigma}_{xx}} \geq 0$$

$$\frac{1}{\pi} \int_0^\pi M_{y \rightarrow x}(\omega) d\omega = \ln \frac{\sigma_{x|x}^2}{\sigma_{x|x,y}^2}$$

- ▶ the integral is the time-domain Granger measure (under a mild condition): $\sigma_{x|x}^2$, $\sigma_{x|x,y}^2$ are the one-step forecast variances of x without and with the past of y

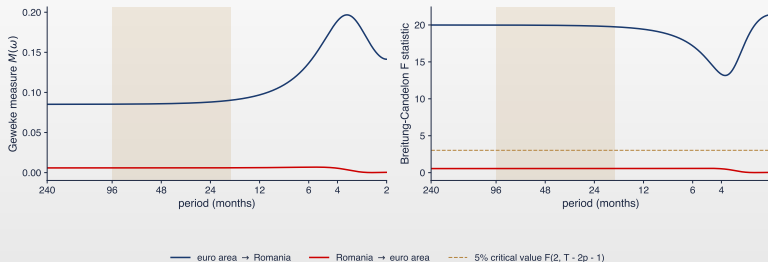
Granger causality by frequency (2/2): the Breitung–Candelon test

- **Breitung and Candelon (2006)**: no causality at frequency ω is two linear restrictions on the coefficients of y in the x equation

$$M_{y \rightarrow x}(\omega) = 0 \iff \sum_{j=1}^p \psi_j \cos(j\omega) = 0 \text{ and } \sum_{j=1}^p \psi_j \sin(j\omega) = 0$$

- ▶ ψ_j : coefficient of y_{t-j} in the x equation; p : VAR order; an $F(2, T - 2p - 1)$ test at each ω , T : number of observations
- ▶ with $p \leq 2$ the two restrictions fix all ψ_j , so the test is the same at every frequency
- BC show that the test extends to cointegrated VARs (long-run causality at $\omega = 0$); pointwise tests across frequencies are not a joint test

Does the euro area cause Romanian industry, and at which frequencies?



- VAR(3) in monthly industrial production growth (order by AIC, at least 3); left: Geweke measures; right: Breitung–Candelon F with its 5% critical value 3.03

Interpreting frequency-domain causality

- Euro area $\rightarrow$ Romania: total Geweke measure 0.146 (integral of the curve: 0.145); Romania $\rightarrow$ euro area: 0.004
- The measure is 0.087 in the business-cycle band and 0.168 at periods under 6 months: past euro-area growth predicts Romanian growth mostly at short horizons
- The BC test rejects at 100% of frequencies for euro area $\rightarrow$ Romania and at no frequency in the other direction
- A small open economy follows its main market, not the reverse; the frequency profile says which kind of shock transmits fastest

Recap: Two series

- ▣ Coherence is an R^2 by frequency; it must be smoothed and tested against $1 - \alpha^{1/(L-1)}$
- ▣ Phase gives leads and lags, modulo 2π , only where coherence is significant; dynamic correlation keeps the sign
- ▣ Geweke decomposes Granger causality by frequency; Breitung–Candelon tests it with two linear restrictions

Band-pass filters

- Ideal filter for periods $[p_l, p_u]$: gain 1 on the frequencies $[\omega_1, \omega_2] = [2\pi/p_u, 2\pi/p_l]$, 0 elsewhere; its weights are infinitely many

$$b_0 = \frac{\omega_2 - \omega_1}{\pi}, \quad b_j = \frac{\sin j\omega_2 - \sin j\omega_1}{\pi j}, \quad j = \pm 1, \pm 2, \dots$$

- [Baxter and King \(1999\)](#): truncate at K lags and add a constant so that $\sum_{j=-K}^K a_j = 0$
 - ▶ a_j : the adjusted weights; the zero sum removes a unit root and a linear trend; symmetric, no phase shift
 - ▶ quarterly recommendation: BK(6, 32, $K = 12$); the price: K observations lost at each end; $a_0 = 0.28$
- [Christiano and Fitzgerald \(2003\)](#): minimise $\mathbb{E}[(y_t^{\text{ideal}} - \hat{y}_t)^2 \mid y_1, \dots, y_n]$ assuming a random walk
 - ▶ y_t^{ideal} : the ideal band-pass component; asymmetric, time-varying weights that use the whole sample: no lost observations, but phase shifts near the ends
- The business-cycle band of 6–32 quarters (1.5–8 years) follows Burns and Mitchell, as adopted by [Baxter and King \(1999\)](#)

The Hodrick–Prescott filter



Edward C. Prescott, 2015

Photo: Narodowy Bank Polski (2015);

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- Hodrick and Prescott (1997): the trend balances fit against smoothness

$$\hat{\tau} = \arg \min_{\tau} \sum_t (y_t - \tau_t)^2 + \lambda \sum_t (\Delta^2 \tau_t)^2 = (I + \lambda D' D)^{-1} y$$

- ▶ τ_t : trend; $\Delta^2 \tau_t$: its second difference; λ : smoothness penalty; D : the second-difference matrix; cycle $y_t - \hat{\tau}_t$

- Infinite-sample gain of the cycle: a high-pass filter

$$G(\omega) = \frac{4\lambda(1 - \cos \omega)^2}{1 + 4\lambda(1 - \cos \omega)^2}$$

- ▶ the Kalman smoother of a local linear trend with signal-to-noise ratio $1/\lambda$ (Chapter 6); $\lambda = 1600$ for quarters; Ravn–Uhlig scaling $\lambda \propto s^4$ (s : observations per quarter): 129 600 for months, 6.25 for years Ravn and Uhlig (2002)

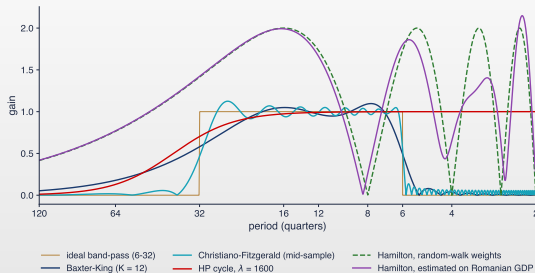
Hamilton's critique

- **Hamilton (2018):** “why you should never use the HP filter”
 - ▶ (1) applied to a random walk, HP produces a cycle with dynamics that are artefacts of the filter [Cogley and Nason \(1995\)](#)
 - ▶ (2) end-of-sample values are very different from the values the same filter gives once more data arrive (one-sided against two-sided)
 - ▶ (3) the λ implied by a statistical model fitted to the data is far from 1600
- Alternative: the **regression filter**, the cycle is the error of an h -step-ahead regression forecast

$$y_{t+h} = \beta_0 + \beta_1 y_t + \cdots + \beta_p y_{t-p+1} + v_{t+h}, \quad \text{cycle}_{t+h} = \hat{v}_{t+h}$$

- ▶ quarterly $h = 8$, $p = 4$; one-sided by construction; for a random walk $v_{t+h} = y_{t+h} - y_t$
- Replies: boosting the HP filter [Phillips and Shi \(2021\)](#); a modified Hamilton filter for real time [Quast and Wolters \(2022\)](#); the debate is about which gain you want

Filters judged by their gain

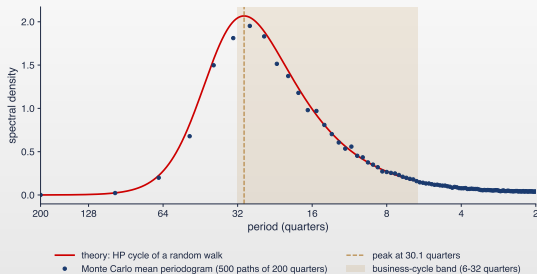


- Gain from the level to the cycle, by period in quarters; Christiano–Fitzgerald: the weights in the middle of a 126-quarter sample; Hamilton: $h = 8$, $p = 4$

Interpreting the gains

- HP is a high-pass filter: gain 0.70 at 32 quarters, 0.49 at 40 and still 0.16 at 60: the “HP cycle” contains cycles longer than 8 years
- BK ripples around 1 (maximum 1.10) and lets in 0.41 of the 40-quarter cycle; CF is closer to the ideal shape in mid-sample
- Hamilton's filter is not a band-pass: gain 2.00 at 16 quarters and 0.00 at 8 (for random-walk weights); the estimated Romanian weights (sum 0.97) give 1.99 and 0.26
- Every filter defines its own cycle: compare cycles only after comparing gains

A cycle made by the filter

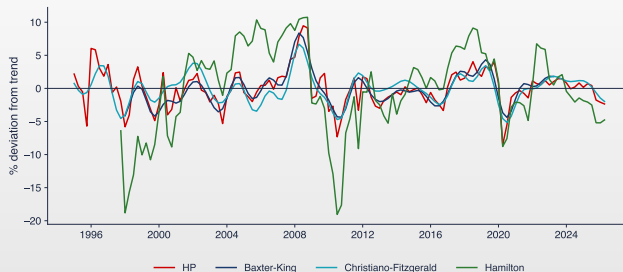


- Spectrum of the HP ($\lambda = 1600$) cycle of a random walk: $G(\omega)^2 / (2\pi \cdot 2(1 - \cos \omega))$, and the average periodogram of HP-filtered simulated random walks

Interpreting the spurious cycle

- A random walk has no cycle, yet its HP cycle peaks at a period of 30.1 quarters (7.5 years); simulation: 28.6
- Derivation (Seminar 11, A1): maximise $u^3/(1 + 4\lambda u^2)^2$ in $u = 1 - \cos \omega$: $u^* = \sqrt{3/(4\lambda)}$; replicates [Cogley and Nason \(1995\)](#)
- The peak sits inside the business-cycle band: “stylised facts” computed from HP cycles can be properties of the filter
- Ravn–Uhlig scaling keeps the peak near 7.5 years at monthly and annual frequency too (Seminar 11, A2)

Romanian business cycles by four filters



- $100 \times \log$ real GDP of Romania, 1995Q1 – 2026Q2; HP ($\lambda = 1600$), BK(6, 32, 12), CF(6, 32, random walk with drift), Hamilton ($h = 8$, $p = 4$)

Interpreting the Romanian cycles

- All four place the 2008 peak (HP 9.5%, BK 8.4%) and the trough of 2010Q3 (HP -7.3% , BK -4.4%) at the same time
- Amplitudes differ: standard deviation HP 2.9, BK 2.5, CF 2.2, Hamilton 6.6; the Hamilton cycle is an 8-quarter forecast error, with a trough of -19.0%
- Correlations: BK–HP 0.84, BK–CF 0.84, HP–Hamilton 0.69, CF–Hamilton 0.49
- Today's position is uncertain: HP -2.4% , CF -2.0% , Hamilton -4.8% in 2026Q2; BK stops at 2023Q2

The end-point problem



- Romanian GDP: the cycle at each date computed with the data available at that date (real time) and with the full sample (final), from 2004Q4

Interpreting the real-time revisions

- HP: root mean square revision 2.8 points, as large as the cycle itself (standard deviation 2.8); correlation of real-time and final cycles 0.57; sign wrong in 38% of quarters
- Hamilton: correlation 0.76, sign wrong in 24% of quarters; but the revision is 4.6 points, because the regression is re-estimated on short samples in a transition economy
- The real-time HP filter saw only a small part of the 2008 boom: the output gap that policy needed was the one it measured worst
- Report real-time properties of any gap estimate; the final estimate is not information that anyone had

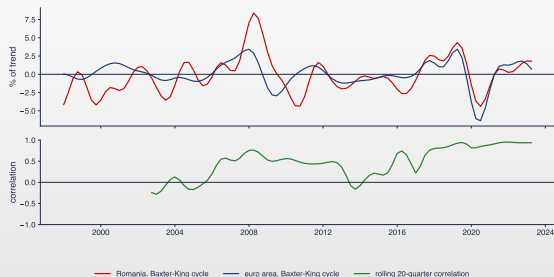
Measuring business-cycle synchronisation

- ▣ Correlation of filtered cycles; it depends on the filter (gain) and on a few large episodes
- ▣ **Concordance** [Harding and Pagan \(2002\)](#): the share of periods in which the two economies are in the same phase

$$C = \frac{1}{T} \sum_t [S_{xt}S_{yt} + (1 - S_{xt})(1 - S_{yt})]$$

- ▶ $S_{xt} = 1$ if x is in expansion (here: cycle above trend), 0 otherwise; under independence
 $\mathbb{E}C = p_x p_y + (1 - p_x)(1 - p_y)$, p_x , p_y : shares of expansion periods
- ▣ Dynamic correlation over the business-cycle band [Croux, Forni and Reichlin \(2001\)](#), which needs no filter
- ▣ Evidence for Central and Eastern Europe: a meta-analysis of 35 publications finds that some countries already had high correlations with the euro area and that the estimation method changes the correlation significantly
[Fidrmuc and Korhonen \(2006\)](#)

Romania and the euro area: synchronisation over time

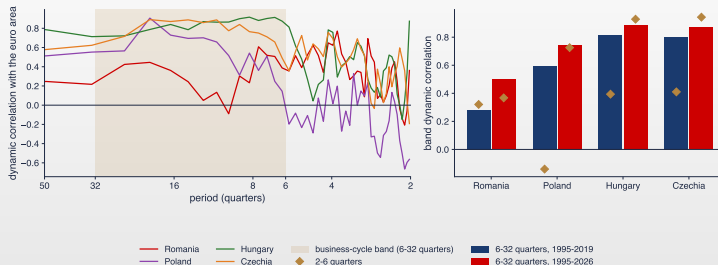


□ Baxter-King cycles of real GDP, 1998Q1 – 2023Q2, and their rolling 20-quarter correlation

Interpreting synchronisation

- ▣ Correlation of the cycles: 0.54 over the whole sample; 0.34 before 2008, 0.47 in 2008–2012, 0.90 in 2013–2019, 0.96 from 2020
- ▣ Concordance 0.75 against 0.50 under independence: the two economies are above or below trend together most of the time
- ▣ The Romanian cycle is larger (standard deviation 2.5 against 1.7): synchronised in timing, not in amplitude
- ▣ The post-2020 correlation is driven by one common shock (the pandemic): a few quarters can make a decade look synchronised

Dynamic correlation with the euro area



- Quarterly GDP growth, multitaper $NW = 3$; left: $\rho(\omega)$ for 1995–2019; right: band averages for 6–32 quarters (bars) and 2–6 quarters (diamonds), without and with 2020–2026

Interpreting the dynamic correlations

- ▣ Before 2020, business-cycle dynamic correlation: Hungary 0.81, Czechia 0.80, Poland 0.60, Romania 0.28
- ▣ Poland's short-run correlation is -0.14 : its growth co-moved with the euro area only through the business cycle
- ▣ With 2020–2026 every number rises (Romania 0.50, Poland 0.75): the pandemic quarter dominates every cross-periodogram
- ▣ Report synchronisation with and without extreme common shocks; robust (trimmed or rank-based) spectra are a research topic

Recap: Filters and the business cycle

- A cycle is defined by a gain: HP is high-pass, BK and CF approximate a band-pass, Hamilton's regression filter is neither
- HP creates a 30-quarter cycle from a random walk and is unreliable in real time
- Romania is synchronised with the euro area in timing since 2010, with a larger amplitude

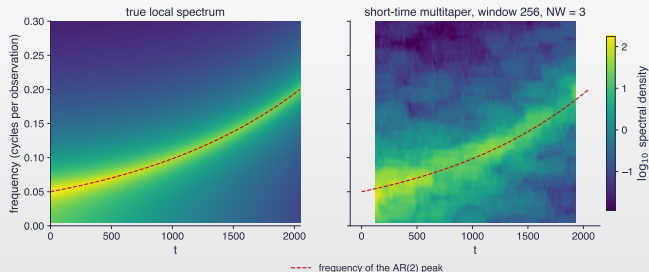
Evolutionary spectra and local stationarity

- ▣ Priestley (1965): Cramér's representation with a slowly changing amplitude

$$X_t = \int A_t(\omega) e^{i\omega t} dZ(\omega), \quad f_t(\omega) = |A_t(\omega)|^2 f(\omega)$$

- ▶ $A_t(\omega)$: time-varying amplitude; $f_t(\omega)$: the **evolutionary spectrum** at time t
- ▣ Dahlhaus (1997): **locally stationary** processes $X_{t,T}$ with $A(t/T, \omega)$ smooth in rescaled time $u = t/T$
 - ▶ T : sample size; a time-varying spectrum $f(u, \omega)$ that can be estimated consistently by a tapered spectrum on a moving window (spectrogram)
- ▣ **Uncertainty principle**: a window of length L resolves frequencies only to about $1/L$; long windows blur time, short windows blur frequency
- ▣ Wavelets (next sections) let the window length change with the frequency: short for high frequencies, long for low ones

A spectrum that drifts



- $X_t = 2r \cos(\theta_t)X_{t-1} - r^2 X_{t-2} + \varepsilon_t$, $r = 0.95$, peak period moving from 20 to 5 over 2048 observations; short-time multitaper on windows of 256 (step 16)

Interpreting the spectrogram

- ▣ The spectrogram follows the moving peak: a single spectrum of the whole sample would show a smeared band from 0.05 to 0.2 cycles
- ▣ Resolution $2NW/L = 0.023$ cycles: fine at high frequencies, coarse for the long periods at the start
- ▣ The first and last 128 observations have no estimate: every time–frequency method loses the edges (the cone of influence of wavelets)
- ▣ Economic examples: changing business-cycle length, the shortening of trading cycles, regime changes in volatility

Singular spectrum analysis in one slide

- Embed $x_1, \dots, x_n$ in the $L \times (n - L + 1)$ trajectory (Hankel) matrix X ; SVD $X = \sum_i \sigma_i u_i v_i'$ Vautard and Ghil (1989)
- Group eigentriples (trend: slowly varying u_i ; oscillations: pairs with equal σ_i and shifted sines) and reconstruct each group by averaging anti-diagonals
- Data-adaptive basis instead of fixed sinusoids or wavelets; no model, no stationarity assumption; inference is less developed
- A code sketch (ssa) is in the Quantlet engine of this chapter; it is a natural benchmark for wavelet multiresolution

Recap: Spectra that change over time

- Priestley and Dahlhaus give a meaning to a time-varying spectrum
- A fixed window trades time against frequency resolution
- Wavelets adapt the window to the frequency; SSA adapts the basis to the data

From sinusoids to wavelets



Ingrid Daubechies, 2005

Photo: own work of the uploader (2005);
public domain; [Wikimedia Commons](#)

- A **wavelet** ψ : a short wave, localised in time and frequency [Grossmann and Morlet \(1984\)](#)
 - ▶ $\int \psi = 0$, $\int \psi^2 = 1$; admissible if $C_\psi = \int |\hat{\psi}(\omega)|^2 / |\omega| d\omega < \infty$, with $\hat{\psi}$ the Fourier transform of ψ
- Continuous wavelet transform (CWT): correlate the series with stretched and shifted copies of ψ

$$W(s, \tau) = \int x(t) \frac{1}{\sqrt{s}} \psi^* \left(\frac{t - \tau}{s} \right) dt$$

- ▶ $s > 0$: scale (inversely related to frequency); τ : location in time; ψ^* : complex conjugate
- Compactly supported orthonormal wavelets [Daubechies \(1988\)](#) and the pyramid algorithm [Mallat \(1989\)](#) give the DWT: n coefficients for n observations; applications: [Crowley \(2007\)](#); [Gençay, Selçuk and Whitcher \(2002\)](#); [Aguilar-Conraria and Soares \(2014\)](#)

DWT, MODWT and multiresolution (1/2)

- Two filters of length L Percival and Walden (2000)

scaling (low-pass): g_l , wavelet (high-pass): $h_l = (-1)^l g_{L-1-l}$

- Haar $L = 2$; Daubechies least-asymmetric LA(8) $L = 8$; level j captures periods $[2^j, 2^{j+1}]$: an **octave** band; J levels plus a smooth

- **MODWT** (maximal overlap DWT): no downsampling, rescaled filters

$$\tilde{h} = h/\sqrt{2}, \quad \tilde{g} = g/\sqrt{2}, \quad \tilde{W}_{j,t} = \sum_l \tilde{h}_{j,l} X_{t-l \bmod n}$$

- $\tilde{h}_{j,l}$: the level- j wavelet filter; $\tilde{W}_{j,t}$: wavelet coefficient of level j at time t ; $\bmod n$: the sample is treated as circular

DWT, MODWT and multiresolution (2/2)

- MODWT properties

- ▶ any sample size, shift-invariant (moving the start date does not change the coefficients), $J \times n$ coefficients
- ▶ energy preserved: $\|X\|^2 = \sum_j \|\tilde{W}_j\|^2 + \|\tilde{V}_J\|^2$, with $\tilde{V}_J$ the scaling (smooth) coefficients

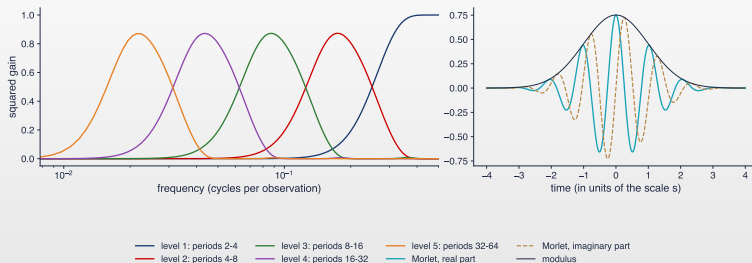
- **Multiresolution analysis (MRA)**: the series is the sum of its details and a smooth

$$X_t = \sum_{j=1}^J \mathcal{D}_{j,t} + \mathcal{S}_{J,t}$$

- ▶ $\mathcal{D}_j$: the inverse MODWT of level j alone; $\mathcal{S}_J$: the smooth; zero-phase, aligned with the data

- **Boundary**: the first $L_j - 1 = (2^j - 1)(L - 1)$ coefficients wrap around the sample (circular filtering) and are excluded from estimation

Wavelet filters and the Morlet wavelet

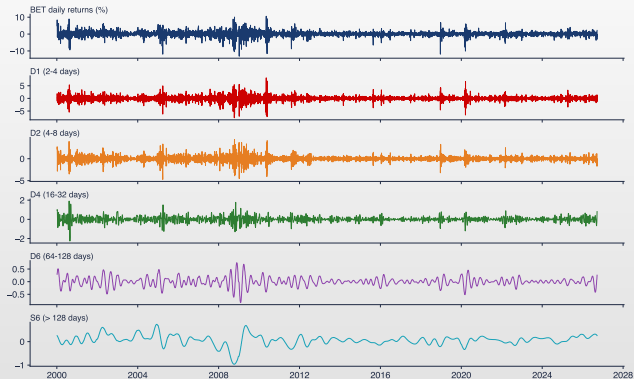


- Left: squared gains of the MODWT LA(8) filters of levels 1–5; right: the Morlet wavelet $\psi_0(\eta) = \pi^{-1/4} e^{i\omega_0 \eta} e^{-\eta^2/2}$, $\omega_0 = 6$

Interpreting the wavelet filters

- ▣ Each MODWT level is an approximate band-pass filter for one octave; the bands overlap, so leakage between neighbouring levels is part of the design
- ▣ The bandwidth grows with frequency: fine frequency resolution for long cycles, fine time resolution for short ones
- ▣ Morlet is complex: its modulus gives amplitude, its argument gives phase; Fourier period = $1.033 \times$ scale for $\omega_0 = 6$
- ▣ Orthonormal filters (MODWT) serve variance decomposition and correlation by scale; the Morlet CWT serves time–frequency maps and phase

The BET return decomposed by scale



- MODWT multiresolution analysis, $LA(8)$, $J = 6$, daily BET returns 2000–2026 (6 670 days); four of the six details and the smooth

Interpreting the multiresolution analysis

- Energy shares of levels 1–6: 43.0, 26.6, 15.0, 6.8, 3.5, 2.3%; white noise would give 50.0, 25.0, 12.5, 6.2, 3.1, 1.6%
- Less energy at the shortest scale and more at the longer ones than white noise: positive autocorrelation of BET returns at short horizons
- Volatility clustering appears at every scale in 2008 and 2020, but the slow components ($\mathcal{D}_6$, $\mathcal{S}_6$) move only in the 2008–2009 crisis
- The details add up exactly to the series: a lossless, zero-phase decomposition

Wavelet variance, correlation and beta by scale (1/2)

- ▣ **Wavelet variance:** the variance of the level- j coefficients; the levels add up to the variance of the series [Percival \(1995\)](#)

$$\nu_X^2(\tau_j) = \text{Var} \tilde{W}_{j,t}, \quad \sum_j \nu_X^2(\tau_j) = \text{Var} X$$

- ▶ $\tau_j = 2^{j-1}$: the scale of level j ; an octave-band version of the spectrum; white noise: $\nu^2(\tau_j) = \sigma^2/2^j$, halving at each level
- ▶ unbiased estimator: average of $\tilde{W}_{j,t}^2$ over the $M_j = n - L_j + 1$ non-boundary coefficients; $\eta \hat{\nu}^2 / \nu^2 \approx \chi_\eta^2$, $\eta = \max(M_j/2^j, 1)$ equivalent degrees of freedom

Wavelet variance, correlation and beta by scale (2/2)

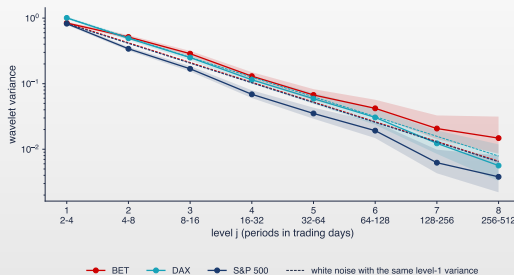
- **Wavelet correlation** Whitcher, Guttorp and Percival (2000) and **wavelet beta** Gençay, Selçuk and Whitcher (2005)

$$\rho_{XY}(\tau_j) = \frac{\text{Cov}(\tilde{W}_j^X, \tilde{W}_j^Y)}{\nu_X(\tau_j)\nu_Y(\tau_j)}$$

$$\beta(\tau_j) = \frac{\text{Cov}(\tilde{W}_j^r, \tilde{W}_j^m)}{\text{Var}\tilde{W}_j^m}$$

- ▶ $\tilde{W}_j^r, \tilde{W}_j^m$: coefficients of an asset return and of the market return; $\beta(\tau_j)$: systematic risk at the investment horizon τ_j
- ▶ 95% interval for the correlation: $\tanh\left(\tanh^{-1}\hat{\rho} \pm z/\sqrt{M_j/2^j - 3}\right)$, $z = 1.96$
- Effective sample size falls by half at each level: long-scale estimates are wide even with 25 years of daily data

Wavelet variance of three equity indices

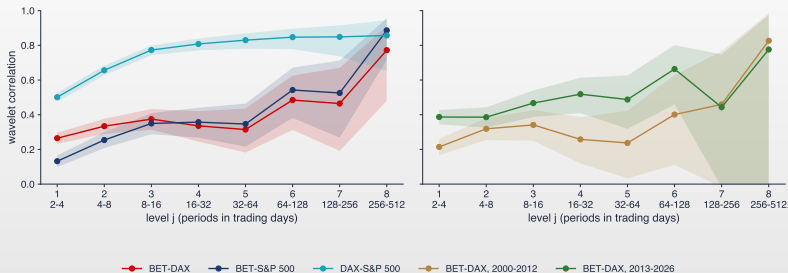


- Daily log returns 2000–2026, MODWT LA(8), levels 1–8 with 95% intervals; dashed: white noise with the same level-1 variance

Interpreting the wavelet variances

- ▣ Ratio of the variance at level 6 (64–128 days) to the white-noise line: BET 1.60, DAX 0.97, S&P 500 0.75
- ▣ BET has more long-horizon variance than its daily variance implies (level 8: 2.25): momentum, a thin market; the S&P 500 has less (0.59): mean reversion of daily moves
- ▣ This is the variance-ratio test of market efficiency, done scale by scale with valid intervals
- ▣ Risk measured on daily data and scaled by $\sqrt{h}$ understates BET risk at long horizons

Co-movement by scale: BET, DAX and S&P 500



□ Daily returns on 6 433 common trading days, MODWT LA(8); left: three pairs, 2000–2026; right: BET–DAX in two halves; 95% Fisher-z intervals

Interpreting the wavelet correlations

- BET–DAX rises from 0.26 (2–4 days) to 0.77 (256–512 days); the daily correlation (0.31) understates long-run integration
- BET–S&P 500 at level 1 is only 0.13, DAX–S&P 500 0.50: New York closes after Europe, so daily co-movement shows up a day late; it vanishes at longer scales
- BET–DAX at 8–16 days: 0.34 in 2000–2012 against 0.47 in 2013–2026 (level 3): integration has increased at short and medium scales
- Wavelet beta of BET on DAX: 0.24 at level 1, 0.56 at level 6: the diversification benefit of the BET shrinks with the horizon (Gençay, Selçuk and Whitcher (2005); Rua and Nunes (2009))

Recap: Wavelets: decomposition by scale

- The MODWT splits variance and covariance into octave bands, for any n and without phase shift
- Wavelet variance is a scale-by-scale variance-ratio test; wavelet correlation and beta show integration by horizon
- Drop boundary coefficients and use equivalent degrees of freedom for intervals

The Morlet transform and its significance (1/2)

- ▣ Torrence and Compo (1998): the CWT computed by FFT on a grid of scales

$$W_n(s) = \sum_k \hat{x}_k \hat{\psi}^*(s\omega_k) e^{i\omega_k n\delta t}, \quad s_j = s_0 2^{j\delta j}$$

- ▶ $\hat{x}_k$: the DFT of the series at frequency ω_k ; $\hat{\psi}$: the Fourier transform of the Morlet wavelet; δt : time step; s_0 : smallest scale; δj : scale step in octaves
- ▶ wavelet power $|W_n(s)|^2$: the variance of the series at time n and scale s
- ▣ **Cone of influence (COI)**: where the wavelet at scale s reaches the edge, within $\sqrt{2}s$ of either end
 - ▶ results inside the COI are biased and are not interpreted

The Morlet transform and its significance (2/2)

- Null of red noise, an AR(1) with lag-1 autocorrelation a : the power is a scaled χ^2_2

$$\frac{|W_n(s)|^2}{\sigma^2} \sim P_k \frac{\chi^2_2}{2}, \quad P_k = \frac{1 - a^2}{1 + a^2 - 2a \cos(2\pi k/N)}$$

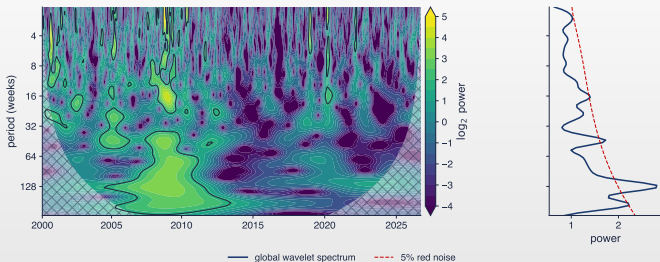
- σ^2 : variance of the series; P_k : the red-noise spectrum at the Fourier index k matching scale s ; N : number of observations

- Global wavelet spectrum: the time average of the power, a smoothed Fourier spectrum

$$\bar{W}^2(s) = \frac{1}{n} \sum_n |W_n(s)|^2, \quad \nu = 2\sqrt{1 + \left(\frac{n\delta t}{2.32s}\right)^2} \text{ degrees of freedom}$$

- Rectify the power by $1/s$ when comparing scales [Liu, Liang and Weisberg \(2007\)](#); the pointwise test runs at thousands of points: expect 5% false patches [Maraun and Kurths \(2004\)](#)

Wavelet power of the BET



- Weekly BET returns (standardised), 1 379 weeks 2000–2026, Morlet $\omega_0 = 6$, $\delta j = 1/12$; black contours: 5% red-noise test; hatched: cone of influence; right: global wavelet spectrum

 `ATS_ch11_wavelet_coherence`

Interpreting the wavelet power

- ▣ Maximum power at a period of 2.6 weeks on 17 October 2008: the Lehman weeks; significant power in 2007–2011 at periods of 32 to 128 weeks
- ▣ Lag-1 autocorrelation 0.04: the red-noise null is almost white; 11% of the reliable area is significant at 5%
- ▣ Power in returns is variance, not a cycle: the map shows when volatility was concentrated at which horizons
- ▣ The global spectrum stays inside the red-noise band: no persistent cycle in BET returns, as efficiency predicts

Wavelet coherence and phase (1/2)

- Cross-wavelet transform and **squared wavelet coherence** Torrence and Webster (1999), Grinsted, Moore and Jevrejeva (2004)

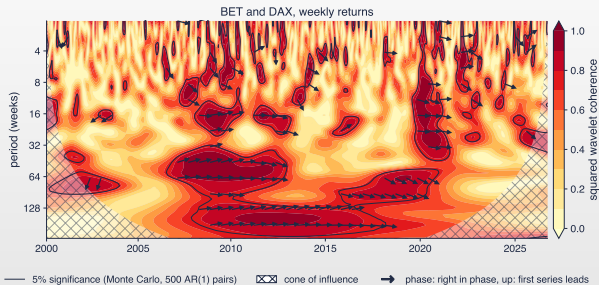
$$W_n^{xy}(s) = W_n^x(s) W_n^{y*}(s)$$
$$R_n^2(s) = \frac{|\mathcal{S}(s^{-1} W_n^{xy}(s))|^2}{\mathcal{S}(s^{-1} |W_n^x(s)|^2) \mathcal{S}(s^{-1} |W_n^y(s)|^2)} \in [0, 1]$$

- ▶ $\mathcal{S}$: smoothing, Gaussian in time (width s) and a boxcar of 0.6 octaves in scale; $R_n^2(s)$: a local R^2 at time n and scale s
- ▶ without smoothing $R^2 \equiv 1$, as for the raw coherence: smoothing is what makes it an estimator

Wavelet coherence and phase (2/2)

- Phase $\arg S(s^{-1}W^{xy})$, shown as arrows
 - ▶ right = in phase, left = anti-phase, up = x leads; lead in time = phase $\times$ period/ 2π
- Significance by **Monte Carlo** [Grinsted, Moore and Jevrejeva \(2004\)](#)
 - ▶ simulate pairs of independent AR(1) series with the data's lag-1 autocorrelations; the 95th percentile of R^2 at each scale is the 5% threshold
- Extensions: partial and multiple wavelet coherence, phase differences with confidence bands [Aguiar-Conraria and Soares \(2014\)](#); monetary policy [Aguiar-Conraria, Azevedo and Soares \(2008\)](#); energy [Vacha and Barunik \(2012\)](#)

BET and DAX in time and frequency

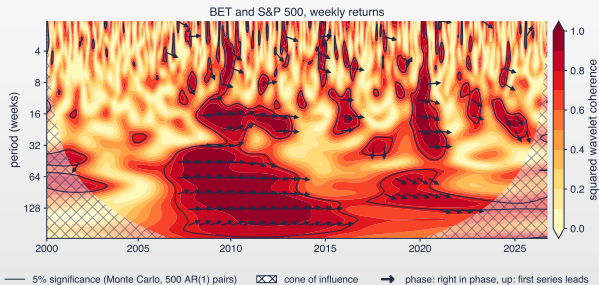


- Weekly returns 2000–2026; contours: 5% significance from 500 AR(1) pairs; arrows only where significant

Interpreting BET–DAX coherence

- 23% of the reliable area is significant (null 95th percentile 7.7%): the co-movement is real, but concentrated
- Average R^2 at 8–64 weeks: 0.37 in 2000–2006, 0.76 in 2008–2009, 0.36 in 2014–2019, 0.77 in 2020–June 2021
- Phase near zero in crises (0.03 rad in 2008–2009), negative before 2007 (−0.97 rad): the DAX led the BET by weeks when the Bucharest market was young
- Crises raise coherence: is it contagion or only higher volatility? Correlations rise mechanically with volatility [Forbes and Rigobon \(2002\)](#), and so does R^2

BET and S&P 500 in time and frequency

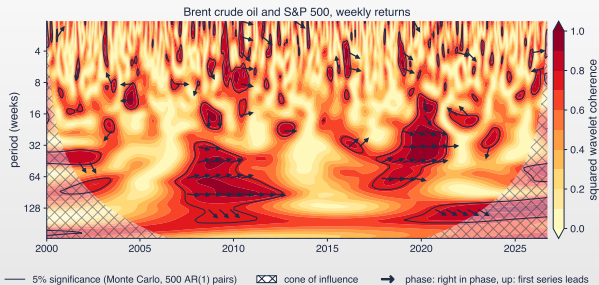


- Weekly returns 2000–2026; Monte Carlo significance as before

Interpreting BET–S&P 500 coherence

- ▣ Significant area 26% (null 95th percentile 7.7%); R^2 at 8–64 weeks 0.76 in 2008–2009 and 0.48 in 2014–2019
- ▣ In 2000–2006 the phase is -1.24 rad: the S&P 500 led by about a fifth of the cycle; in crises the markets move in phase
- ▣ Weekly data remove the one-day lag of the daily wavelet correlation: frequency choice matters for lead–lag claims
- ▣ Our picture matches the international evidence: co-movement is strongest at low frequencies and in crises [Rua and Nunes \(2009\)](#)

Oil and stocks

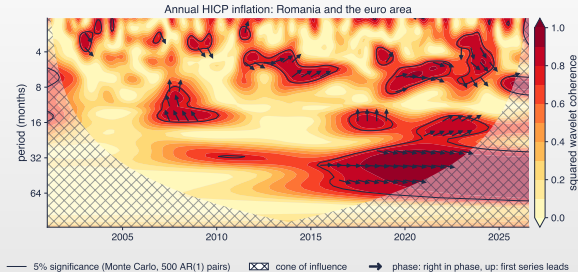


- Weekly returns of Brent crude oil (FRED) and the S&P 500, 2000–2026; Monte Carlo significance from 500 AR(1) pairs

Interpreting oil–stock coherence

- Significant area 15% (null 95th percentile 7.1%): oil and stocks are linked only in episodes
- Before 2007: R^2 0.41 with phase -1.86 rad, neither in phase nor in anti-phase: a weak lead–lag link; 2008–2010: 0.70, in phase (-0.09 rad): a common demand shock
- 2020–2021: 0.58 in phase; 2022–2023: phase -2.31 rad, closer to anti-phase: the energy–price shock of the war in Ukraine
- The sign of the oil–stock link depends on the type of shock [Kilian and Park \(2009\)](#); wavelet phase shows the switch without identifying it

Inflation in Romania and the euro area



- Annual HICP inflation (monthly), January 2001 – August 2026; Monte Carlo significance against AR(1) pairs with lag-1 autocorrelation 0.97

Interpreting inflation co-movement

- Romania–euro area: R^2 at 24–64 months 0.23 in 2002–2012, 0.85 from 2015; significant area 17% (null 95th percentile 8.7%)
- Poland: 0.29 and 0.91; simple correlations 0.34 (Romania) and 0.86 (Poland): Romanian disinflation to 2012 was domestic, the cycle since 2015 is common
- 2021–2024 surge, 12–48 months: R^2 0.78, phase 0.14 rad: no detectable lead, a common energy shock
- Annual rates overlap twelve months: short periods are smoothed away by construction; only cycles longer than a year are informative

Recap: Wavelet coherence

- Wavelet coherence is a smoothed, time-local coherence; without smoothing it is identically one
- Test it by Monte Carlo against AR(1) pairs; read phase only inside significant regions and outside the cone of influence
- Equity co-movement peaks in crises and at long horizons; oil–stock phase flips with the type of shock

An open question

- Is the rise of stock-market coherence in crises contagion, or the mechanical effect of higher volatility on a time-local R^2 ?
 - ▶ formal: H_0 : the wavelet coherence of BET and DAX in the pre-registered crisis windows equals its value under a model with the same time-varying volatilities and constant dependence
 - ▶ falsified if the crisis coherence exceeds the 95th percentile of a bootstrap from a DCC-GARCH with constant correlation ([Chapter 8](#)), at pre-registered scales
- Why it matters: diversification fails when it is needed; the Forbes–Rigobon correction exists for correlations, not for wavelet coherence
 - ▶ literature to start from: [Forbes and Rigobon \(2002\)](#), [Rua and Nunes \(2009\)](#), [Maraun and Kurths \(2004\)](#), [Schulte \(2016\)](#), [Aguiar-Conraria and Soares \(2014\)](#)

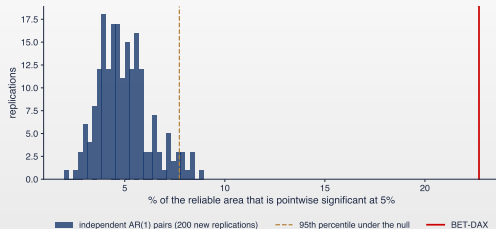
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
 - ▶ **literature:** List peer-reviewed papers since 2009 that test wavelet coherence between stock markets against a volatility-matched null; give DOIs. Then check every DOI on Crossref
 - ▶ **hypothesis:** Derive the bias of the smoothed wavelet coherence when both series have a common GARCH volatility and constant correlation.
 - ▶ **code and replication:** ask for the Grinsted smoothing operator, then reproduce a known case first (two AR(1) series: 5% significant area)
 - ▶ **critique:** Act as a hostile referee: list the ways a significant wavelet-coherence island can arise without any change in dependence.
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

What the human checks

- ▣ Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- ▣ Conventions: angular frequency or cycles, 2π factors, period = $1.033\times$ scale for Morlet, the sign convention of the phase
- ▣ Coherence is smoothed; its threshold and the Monte Carlo null are stated; nothing is read inside the cone of influence
- ▣ Pointwise significance is not areawise significance: count how many islands the null produces
- ▣ Filters: gain stated; real-time and final cycles separated; results with and without extreme common shocks

Mini-case: how many islands does chance produce?



- Share of the reliable time–period area of a wavelet-coherence map that is pointwise significant at 5%: 200 new pairs of independent AR(1) series with the BET and DAX autocorrelations, and the observed BET–DAX map
- Null: mean 5.0%, 95th percentile 7.7%, maximum 9.0%; 100% of null maps contain significant islands; BET–DAX: 22.7%. An AI summary that reads every island as a contagion episode is wrong; the overall excess is real

Project idea

- ▣ **Contagion or volatility? Wavelet coherence of Central and Eastern European stock markets under a volatility-matched null**
 - ▶ replicate: the Monte Carlo test of [Grinsted, Moore and Jevrejeva \(2004\)](#) and the international co-movement maps of [Rua and Nunes \(2009\)](#) on BET, WIG20, BUX, PX against the DAX
 - ▶ extend: a DCC-GARCH bootstrap null (constant dependence, time-varying volatility); areawise tests [Schulte \(2016\)](#); partial coherence controlling for the S&P 500 [Aguilar-Conraria and Soares \(2014\)](#)
 - ▶ pre-register: markets, sampling frequency, crisis windows, scales, the null model and the number of replications
- ▣ Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- ▣ The spectrum decomposes variance by frequency; filters act on it through their squared gain
- ▣ Estimation is a bias–variance choice; multitaper gives low leakage, stable variance and honest bands
- ▣ Coherence, phase, dynamic correlation and Breitung–Candelon tests describe co-movement and causality by frequency
- ▣ A business cycle is whatever the filter's gain lets through; HP adds spurious cycles and fails in real time
- ▣ Wavelets localise variance and co-movement in time and scale; significance needs Monte Carlo and areawise thinking

Self-assessment

Questions

- Why is the periodogram inconsistent although it is asymptotically unbiased?
- How does the optimal bandwidth grow with n for a kernel with $q = 2$?
- Why is raw (unsmoothed) coherence always equal to one?
- Where does the peak of the HP cycle of a random walk come from?
- Why is the MODWT preferred to the DWT for wavelet variance?

Next:

- Machine learning and deep learning for time series
- global models, leakage-free validation, deep architectures

Appendix: the distribution of the periodogram

- Gaussian white noise: $a_j = \sqrt{2/n} \sum_t x_t \cos(\omega_j t)$, $b_j = \sqrt{2/n} \sum_t x_t \sin(\omega_j t)$ are i.i.d. $N(0, \sigma^2)$ for $0 < \omega_j < \pi$ (orthogonality of the Fourier basis)
- $I(\omega_j) = (a_j^2 + b_j^2)/(4\pi) = \frac{\sigma^2}{2\pi} \cdot \frac{\chi_2^2}{2}$: mean $f = \sigma^2/(2\pi)$, variance f^2 , whatever n
- Linear process: $d_X(\omega_j) = \Psi(e^{-i\omega_j})d_\varepsilon(\omega_j) + o_p(1)$, so $I_X(\omega_j) \approx |\Psi(e^{-i\omega_j})|^2 I_\varepsilon(\omega_j) = f(\omega_j)\chi_2^2/2$
- Averaging L such ordinates: mean f , variance f^2/L ; this is the logic of every smoothed and multitaper estimator

Appendix: two filter facts

- ▣ **Coherence is filter-invariant:** $\tilde{x} = A(L)x$, $\tilde{y} = B(L)y$ give $f_{\tilde{x}\tilde{y}} = A(\omega)\overline{B(\omega)}f_{xy}$, $f_{\tilde{x}} = |A|^2f_x$, $f_{\tilde{y}} = |B|^2f_y$, so $\kappa_{\tilde{x}\tilde{y}}^2 = \kappa_{xy}^2$ where $A, B \neq 0$
- ▣ The phase changes by $\arg A - \arg B$: prewhitening with different filters shifts the phase; use the same filter on both series
- ▣ **HP cycle of a random walk:** $\Delta y = \varepsilon$, so $f_c(\omega) = G(\omega)^2\sigma^2/(2\pi \cdot 2(1 - \cos\omega))$; with $u = 1 - \cos\omega$ this is proportional to $u^3/(1 + 4\lambda u^2)^2$
- ▣ Setting the derivative to zero: $3(1 + 4\lambda u^2) = 16\lambda u^2$, so $u^* = \sqrt{3/(4\lambda)}$ and the period $2\pi/\arccos(1 - u^*)$

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